# The Impact of User Interaction History on Collaborative, Content-Based, and Hybrid Recommendation Performance

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**Figure 1: Seattle Mariners at Spring Training, 2010.**

## Abstract

This study examines how user interaction history relates to recommendation performance across user activity groups. Using the Amazon Reviews 2023 Musical Instruments dataset with the provided temporal training and warm-validation split, we compare four recommendation approaches: a popularity baseline, a TruncatedSVD-based latent-factor model, TF-IDF content-based filtering, and a hybrid model combining the two latter approaches. Users are grouped by the number of training interactions into Low (1–5), Medium (6–8), and High (9 or more) activity groups. All approaches are evaluated using Recall@10 and NDCG@10 under a shared evaluation protocol. In the current experiments, the hybrid model achieves the highest overall Recall@10 (0.014567), while the popularity baseline achieves the highest overall NDCG@10 (0.009222). Group-level results show that performance varies with user activity and evaluation metric. These findings provide an empirical comparison of recommendation approaches under different levels of user interaction history. The results are limited to the current warm-validation setting and do not establish statistical significance.

## CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

## Keywords

recommender systems, user-history sparsity, collaborative filtering, TF-IDF, hybrid recommendation

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and supplementary experiments belong in the reproducibility package.

# 1 Introduction

Recommendation systems help users discover relevant items by ranking alternatives according to their likely interests. Their performance depends partly on the information available about each user. Users with extensive interaction histories provide more evidence of their preferences, whereas users with only a few interactions provide less information for estimating their interests. This limitation is commonly associated with user-history sparsity and may affect recommendation approaches differently.

Collaborative filtering uses patterns in user–item interactions to estimate preferences. In this study, the collaborative baseline uses Truncated Singular Value Decomposition (TruncatedSVD) to obtain latent representations from the user–item rating matrix. Content-based filtering instead represents items using their textual attributes and recommends items according to their similarity to a user's positive interaction history. The study implements this approach using Term Frequency–Inverse Document Frequency (TF-IDF). A hybrid model combines the scores from the latent-factor and content-based components, while a popularity-based method provides a reference baseline.

The effect of interaction history is important because the four approaches rely on different sources of information. Popularity uses aggregate positive interactions, the latent-factor approach learns from the user–item interaction matrix, and TF-IDF content-based filtering relies on item metadata and a user's positive history. The hybrid approach combines interaction-based and content-based signals. Comparing these methods across users with different amounts of training history can help identify whether one approach performs better than another under the evaluated conditions.

This study uses the Amazon Reviews 2023 Musical Instruments dataset with the provided temporal data split. The training data contain 427,957 interactions from 54,483 users. The experiments use the provided warm-validation data and retain the original split rather than shuffling or reallocating interactions. Users are grouped by the number of training interactions: Low (1–5), Medium (6–8), and High (9 or more). Recommendation performance is measured using Recall@10 and Normalized Discounted Cumulative Gain at 10 (NDCG@10) under a shared evaluation procedure.

The study addresses the following research questions:

- RQ1: How does the number of user interactions relate to the performance of collaborative, content-based, and hybrid recommendation approaches?
- RQ2: Which recommendation approach achieves the most consistent performance across user groups with different levels of interaction history?

The contribution of this study is an empirical comparison of popularity-based, TruncatedSVD-based, TF-IDF content-based, and hybrid recommendation approaches across user-history groups. Rather than proposing a new recommendation algorithm, the study examines differences in Recall@10 and NDCG@10 across the defined activity groups using the same warm-validation protocol.

# 2 Related Work

## 2.1 Recommendation Approaches

Recommendation systems commonly use collaborative filtering, content-based filtering, and hybrid approaches to generate personalized recommendations. Collaborative filtering identifies preference patterns from user–item interactions, while content-based filtering uses item attributes to identify items similar to those previously preferred by a user [1, 2]. Hybrid recommendation combines different recommendation approaches to use complementary sources of information [3]. These approaches provide the methodological basis for comparing recommendation performance under different levels of user interaction history.

## 2.2 User Interaction History and Recommendation Evaluation

The amount of interaction history available for each user can affect how effectively a recommendation model estimates preferences. Collaborative approaches depend on interaction data, whereas content-based approaches can use item descriptions and other textual attributes to represent user interests. A hybrid approach combines these two sources of evidence. However, the relative performance of these methods cannot be determined from their design alone and should be examined through a shared evaluation protocol.

Recommendation evaluation requires consistent metrics and data partitions so that performance comparisons are meaningful [4]. In this study, Recall@10 measures the proportion of relevant validation items retrieved among the top ten recommendations, while NDCG@10 evaluates the ranking of relevant items within those recommendations. The same evaluation procedure is applied to all four methods.

## 2.3 Position of This Study

This study compares a popularity-based baseline, a TruncatedSVD-based latent-factor model, TF-IDF content-based filtering, and a hybrid model combining the latent-factor and content-based scores. Users are grouped by their number of training interactions into Low (1–5), Medium (6–8), and High (9 or more) activity groups. The comparison investigates how recommendation performance differs across these groups using the provided warm-validation setting. Thus, the study focuses on an empirical comparison under a shared protocol rather than proposing a new recommendation algorithm.

# 3 Dataset and Problem Formulation

## 3.1 Dataset Description

This study uses the Amazon Reviews 2023 Musical Instruments This study uses the Amazon Reviews 2023 Musical Instruments dataset with the provided 5-Core subset and absolute-timestamp split. The training set contains 427,957 interactions from 54,483 users. The full validation set contains 41,284 interactions, while the warm-validation subset used in the experiments contains 30,360 interactions from 11,659 evaluable users. The test set contains 42,595 interactions and is reserved for final evaluation. It is not used for model selection or parameter tuning.

**Table 1: Recommended minimum manuscript allocation. These are minimum targets, not rigid limits. Additional space should primarily go to Method and Results and Discussion. Delete this table before final submission.**

| Section | Minimum | Section | Minimum |
|---|---|---|---|
| Abstract | 0.25 page | Results and Discussion | 1.50 pages |
| Introduction | 0.75 page | Limitations / Responsible Recommendation | 0.25 page |
| Related Work | 0.50 page | Conclusion | 0.25 page |
| Dataset and Problem Formulation | 0.50 page | AI Usage Declaration | 0.10 page |
| Method | 0.90 page | Contribution | 0.15 page |
| Experimental Setup | 0.75 page | References | 0.75 page |

Each interaction contains a user ID, item ID, rating, and timestamp. Ratings of 4 or 5 are treated as positive interactions when constructing the popularity ranking and the positive user profiles for content-based filtering. Item metadata, including title, main category, categories, features, description, and basic content text, is used to represent item characteristics in the TF-IDF model.

The experiments use the provided candidate-item set, which contains 22,753 items. Items already observed in a user's training history are excluded from that user's recommendations. Users are grouped according to their number of training interactions: Low (1–5 interactions), Medium (6–8 interactions), and High (9 or more interactions). These fixed groups allow recommendation performance to be compared across different levels of user interaction history.

## 3.2 Problem Formulation

The task is to rank candidate items for each evaluable user based on the user's training interaction history. Four recommendation approaches are compared: a popularity-based baseline, a Truncated Singular Value Decomposition (TruncatedSVD) latent-factor baseline, a TF-IDF content-based filtering baseline, and a hybrid model combining the latent-factor and content-based scores. All models use the same training data, candidate-item set, warm-validation subset, and evaluation procedure to support a consistent comparison.

For the popularity baseline, candidate items are ranked by the number of positive training interactions, defined as ratings of 4 or 5. The collaborative baseline estimates user-item preference scores using TruncatedSVD applied to the training interaction matrix. The content-based baseline represents item metadata using TF-IDF and ranks candidates by cosine similarity to each user's profile, constructed from positively rated training items. The hybrid model combines the normalized TruncatedSVD and TF-IDF scores with equal weights of 0.5.

The study evaluates recommendation quality using Recall@10 and Normalized Discounted Cumulative Gain at 10 (NDCG@10). Recall@10 measures the proportion of relevant validation items retrieved in the top ten recommendations, while NDCG@10 accounts for the positions of relevant items in the ranking. For the group analysis, users are categorized by the number of training interactions into Low (1–5), Medium (6–8), and High (9 or more). The comparison examines how the four approaches perform overall and across these user-activity groups. The test set is reserved for final evaluation and is not used in the current warm-validation comparison.

**Table 2: Interaction Data Used in the Experiments**

| Split | Interactions | Purpose |
|---|---|---|
| Training | 427,957 | Model fitting |
| Full validation | 41,284 | Reference split |
| Warm validation | 30,360 | Model evaluation |
| Test | 42,595 | Final evaluation only |

The dataset consists of training, validation, and test interactions from the Amazon Reviews 2023 Musical Instruments dataset. As shown in Table 2, the training set contains 427,957 interactions and is used to fit the recommendation models. The full validation set contains 41,284 interactions, of which 30,360 interactions belong to the warm-validation subset used for model evaluation. The test set contains 42,595 interactions and is reserved for final evaluation only. The experiments use the provided warm-validation data to compare the performance of the popularity-based, TruncatedSVD-based, TF-IDF content-based, and hybrid recommendation approaches. The test set is not used for model selection or parameter tuning.

# 4 Method

## 4.1 Popularity Baseline

The popularity baseline ranks candidate items according to the number of positive interactions in the training data, where an interaction is considered positive if its rating is at least 4. For each candidate item, the number of positive interactions is counted. Candidate items are then sorted in descending order of positive interaction counts, with item identifiers in ascending order used to break ties. The resulting global ranking is shared across users and is saved for evaluation. During evaluation, the ranking is assessed using the common evaluation procedure applied to all recommendation models.

## 4.2 Collaborative Filtering Baseline

The collaborative filtering baseline uses Truncated Singular Value Decomposition (TruncatedSVD) as a latent-factor approximation. The training interactions are aggregated by taking the mean rating for each user–item pair and represented as a sparse user–item matrix. TruncatedSVD is applied to obtain user and item latent representations. The number of components is set to the minimum of 32 and one less than the smaller matrix dimension, with seven iterations and a random state of 42.

For each evaluation user, candidate-item scores are computed as the dot product between the user's latent representation and each candidate item's latent representation. Candidate items already observed by the user in the training data are excluded. The remaining items are ranked by descending predicted score, with item identifiers in ascending order used to break ties. Up to ten highest-ranked items are returned for evaluation.

### 4.3 TF-IDF Content-Based Filtering

The content-based filtering baseline represents items using their textual metadata. Available fields, including title, main category, categories, features, description, and basic content text, are concatenated into a single text representation. TF-IDF vectors are constructed using up to 100,000 features, English stop-word removal, a minimum document frequency of two, and unigram and bigram features. Duplicate item metadata entries are removed before vectorization.

For each evaluation user, a profile is constructed by averaging the TF-IDF vectors of items rated at least four in the training data. The resulting profile is normalized, and cosine similarity is computed between the user profile and each candidate item's TF-IDF vector. Candidate items are ranked in descending order of similarity, with item identifiers in ascending order used to break ties. Items already observed in the user's training history are excluded, and up to ten remaining items are recommended. Users without positive training items that have available metadata receive an empty recommendation list.

### 4.4 Hybrid Recommendation Model

The hybrid model combines the TruncatedSVD-based latent-factor approach and TF-IDF content-based filtering. The latent-factor component uses the sparse user–item rating matrix constructed from the training data, while the content-based component uses item metadata, including titles, categories, features, descriptions, and available basic content text.

For each evaluation user, the latent-factor score for each candidate item is computed as the dot product of the corresponding user and item representations. The content-based score is calculated using cosine similarity between the user's profile and the candidate item's TF-IDF vector. The user profile is obtained by averaging the TF-IDF vectors of items rated at least four in the training data.

Before combination, each component's scores are independently normalized using min–max normalization. The hybrid score is calculated as

$$s_{\text{hybrid}}(u, i) = 0.5\hat{s}_{\text{MF}}(u, i) + 0.5\hat{s}_{\text{CBF}}(u, i), \tag{1}$$

where $u$ denotes a user, $i$ denotes a candidate item, and the normalized component scores are denoted by $\hat{s}_{\text{MF}}$ and $\hat{s}_{\text{CBF}}$. Thus, both components receive equal initial weights.

Candidate items already observed by a user in the training data are excluded. The remaining items are ranked by descending hybrid score, with item identifiers in ascending order used to break ties. The ten highest-ranked items are selected for recommendation, and the resulting rankings are evaluated using the same evaluation procedure as the other baselines.

## 5 Experimental Setup

*Template guidance (delete before submission).* Minimum target: **0.75 page**. Make the comparison reproducible and fair. State explicitly that training is used for model fitting, validation for model selection and hyperparameter tuning, and test only for final evaluation after the configuration is frozen. Define evaluable users, candidate items, removal of seen training items, Recall@10 and NDCG@10, any track-specific metric/analysis, hyperparameter search space, final validation-selected configuration, random seed(s), and important software/library versions. All compared methods must use the same candidate protocol where the comparison requires it.

*Replace this guidance with your Experimental Setup.*

## 6 Results and Discussion

*Template guidance (delete before submission).* Minimum target: **1.50 pages**; this should normally be the largest section. Include: (1) the main baseline comparison on the final test set; (2) at least one track-specific analysis that directly answers the RQ; (3) interpretation of why the result may have occurred; (4) failure/error analysis; and (5) an explicit answer to each RQ. Do not merely state which number is highest.

**Table 3: Example of a clear final-results table. Values are placeholders only.**

| Method | Recall@10 ↑ | NDCG@10 ↑ | Coverage ↑ |
|---|---|---|---|
| Popularity | 0.121 | 0.084 | 0.032 |
| MF | 0.208 | 0.157 | 0.184 |
| Hybrid | **0.231** | **0.176** | 0.207 |
| Track method | 0.226 | 0.172 | **0.254** |

*Good table example (delete before submission).* A good table identifies the methods, metrics, direction of improvement, split or context in the caption, and uses sensible precision. Boldface has an explicit meaning rather than being decorative.

**Table 4: Example of a poor table that should not appear in the manuscript.**

| Alg1 | Alg2 | Alg3 |
|---|---|---|
| 0.473284758 | 0.487991123 | 0.486882772 |

*Bad table example (delete before submission).* This table is poor because the metric, split, model meaning, and interpretation are unknown, while the excessive decimal precision adds no value.

*Good figure example (delete before submission).* A good figure directly supports an RQ. Its axes are labeled, method names are meaningful, the caption explains the experimental context, and the figure remains readable when printed at column width.

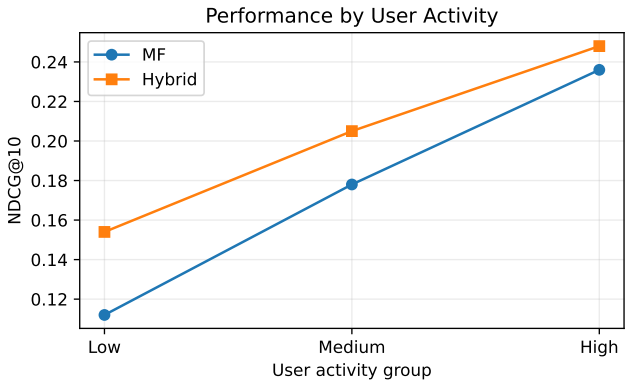



**Figure 2: Example of a useful research figure: NDCG@10 across user-activity groups for two clearly named methods. The axes, metric, groups, and legend are interpretable at ACM column width. Placeholder values are used.**

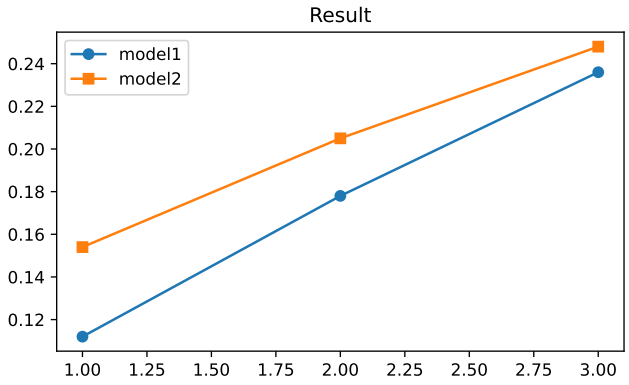



**Figure 3: Example of a poor research figure. Generic method names, missing axis labels, and the vague title "Result" make the plot difficult to interpret.**

*Bad figure example (delete before submission).* Also avoid screenshots of Jupyter/Colab, screenshots of code or CSV files, decorative AI-generated images, unreadable plots with too many series, and any figure that does not provide evidence for a research question. For a 6–8 page paper, a useful visual budget is normally **2–3 tables and 1–2 figures**.

*Replace this guidance and all example tables/figures with your Results and Discussion.*

## 7 Limitations / Responsible Recommendation Considerations

*Template guidance (delete before submission).* Minimum target: **0.25 page**. Discuss limitations that affect validity or generalizability, such as using only one Amazon category, offline evaluation, the rating $\geq 4$ relevance convention, metadata quality, or track-specific trade-offs. Also discuss responsible-recommendation implications relevant to the track, such as popularity concentration, diversity, calibration, or reinforcing past preferences. "Our laptop was slow" is not a scientific limitation unless it materially changed the experimental design.

*Replace this guidance with your Limitations / Responsible Recommendation Considerations.*

## 8 Conclusion

*Template guidance (delete before submission).* Minimum target: **0.25 page**. In one or two compact paragraphs, restate the objective, answer the RQ(s) directly, state the most important finding, and give one realistic future-work direction. Do not introduce a new experiment or a new claim that was not supported earlier in the paper.

*Replace this guidance with your Conclusion.*

## 9 AI Usage Declaration

*Template guidance (delete before submission).* Minimum target: about **0.10 page**. State factually whether and how generative AI tools were used, for example for brainstorming, language editing, coding assistance, debugging, or not at all. The authors remain responsible for originality, correctness, citations, analysis, code, and experimental results.

*Example wording (delete before submission). Generative AI tools were used for language editing and debugging assistance. They were not used to fabricate experimental results or references. All AI-assisted outputs were reviewed and verified by the authors.*

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## 10 Contribution

*Template guidance (delete before submission).* Minimum target: about **0.15 page**. State the contribution of each group member using concrete roles such as data preparation, baseline implementation, track-specific method, experimentation, evaluation, visualization, literature review, manuscript drafting/revision, reproducibility, and presentation. Avoid "all members contributed equally" unless it is accurate and individual roles can still be explained during Week 15 questioning.

*Example wording (delete before submission). Student A: data preparation and MF implementation; Student B: CBF and visualization; Student C: track-specific experiment and evaluation; Student D: literature review, manuscript revision, and reproducibility checks.*

*Replace this guidance with the contribution statement for your group.*

*References guidance (delete before submission).* Minimum target: approximately **0.75 page**. For this course manuscript, aim for roughly **8–15 strong references**. Prefer peer-reviewed or authoritative scholarly sources and cite them because they support a claim. Avoid using blogs, Medium posts, random tutorials, or ChatGPT as scholarly references for algorithmic/scientific claims. Use BibTeX and include DOI when available. The Rosni Lumbantoruan examples cited in the Related Work guidance above are stored in `sample-base.bib`; delete any example citation you do not actually use in the final manuscript.

## References

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